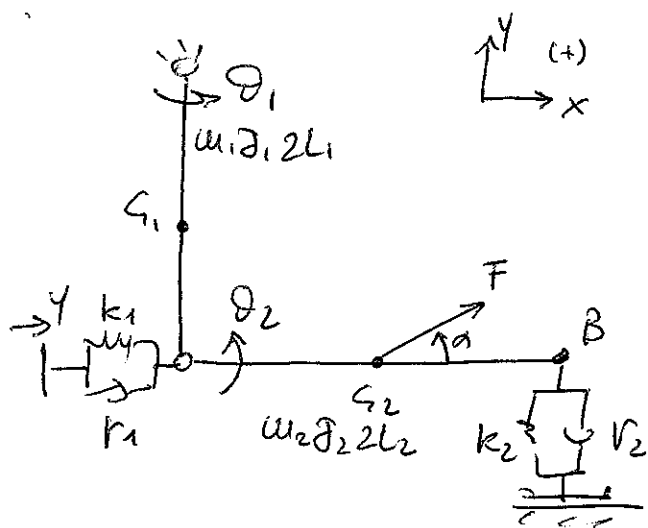


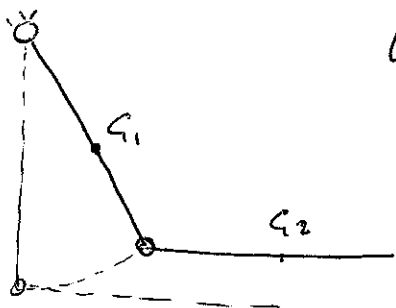


$$\underline{x} = \begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix}$$

$\theta_1, \theta_2 = \text{rotation absolute}$

$$E_c = \frac{1}{2} m_1 v_{G1}^2 + \frac{1}{2} J_1 \dot{\theta}_1^2 + \frac{1}{2} m_2 v_{G2}^2 + \frac{1}{2} J_2 \dot{\theta}_2^2$$

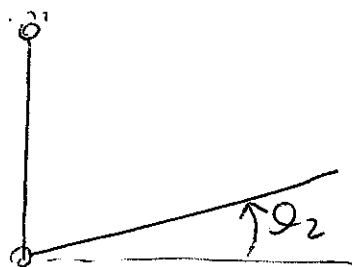
• $\theta_1 \neq 0 ; \theta_2 = 0$



cinematica lineare rispetto a:

$$\begin{cases} x_{G1} = L_1 \theta_1 \\ y_{G1} = 0 \end{cases} \quad \begin{cases} x_{G2} = 2L_1 \theta_1 \\ y_{G2} = 0 \end{cases}$$

• $\theta_1 = 0 ; \theta_2 \neq 0$



$$\begin{aligned} \rightarrow x_{G1} &= y_{G1} = 0 \\ x_{G2} &= 0 \\ y_{G2} &= L_2 \theta_2 \end{aligned}$$

$$\rightarrow \begin{cases} x_{G1} = L_1 \theta_1 \\ y_{G1} = 0 \\ x_{G2} = 2L_1 \theta_1 \\ y_{G2} = L_2 \theta_2 \end{cases}$$

$m_F = \text{diag}(m_1, J_1, m_2, J_2)$

$$[A_m] = \begin{bmatrix} L_1 & 0 \\ 1 & 0 \\ 2L_1 & 0 \\ 0 & L_2 \\ 0 & 1 \end{bmatrix}$$

$$V = \frac{1}{2} k_1 \Delta L_1^2 + \frac{1}{2} k_2 \Delta L_2^2$$

$$\begin{cases} \Delta L_1 = 2L_1 \sin \vartheta_1 \\ \Delta L_2 = 2L_1 (1 - \cos \vartheta_1) + 2L_2 \sin \vartheta_2 \end{cases}$$

$$\frac{\partial \Delta L_1}{\partial \vartheta_1} = 2L_1 \cos \vartheta_1, \quad \frac{\partial^2 \Delta L_1}{\partial \vartheta_1^2} = -2L_1 \sin \vartheta_1 \rightarrow \left. \frac{\partial^2 \Delta L_1}{\partial \vartheta_1^2} \right|_0 = 0$$

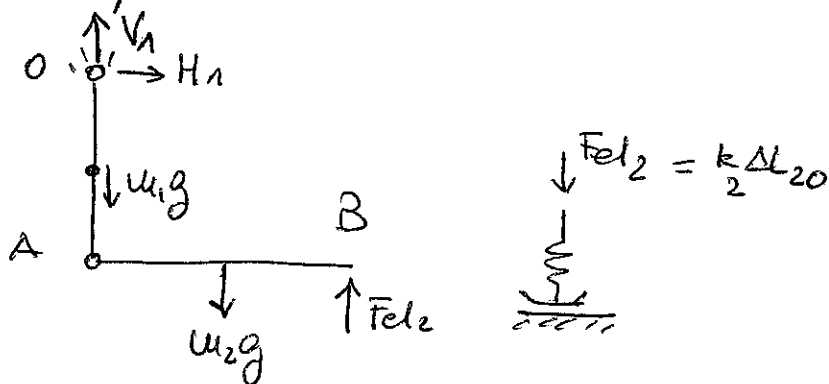
$$\frac{\partial \Delta L_2}{\partial \vartheta_1} = 2L_1 \sin \vartheta_1, \quad \frac{\partial^2 \Delta L_2}{\partial \vartheta_1^2} = 2L_1 \cos \vartheta_1 \rightarrow \left. \frac{\partial^2 \Delta L_2}{\partial \vartheta_1^2} \right|_0 = 2L_1$$

$$\frac{\partial \Delta L_2}{\partial \vartheta_2} = 2L_2 \cos \vartheta_2, \quad \frac{\partial^2 \Delta L_2}{\partial \vartheta_2^2} = -2L_2 \sin \vartheta_2 \rightarrow \left. \frac{\partial^2 \Delta L_2}{\partial \vartheta_2^2} \right|_0 = 0$$

$$k_F = \text{diag}(k_1, k_2) \quad [A_{EI}] = \begin{bmatrix} 2L_1 & 0 \\ 0 & 2L_2 \end{bmatrix}$$

- la molla k_1 non è pretesa e $\frac{\partial^2 \Delta L_1}{\partial \vartheta_1^2} = 0 \rightarrow k_{II}^{(1)} = 0$
- la molla k_2 è pretesa e $\frac{\partial^2 \Delta L_2}{\partial \vartheta_2^2} = 2L_1 \rightarrow k_{II}^{(2)} \neq 0$

calcolo pretesa:



$$\sum M_A^{AB} = 0 \rightarrow u_2 g L/2 - F_{el2} 2L/2 = 0 \rightarrow F_{el2} = \frac{u_2 g}{2} \rightarrow$$

$$\rightarrow \Delta L_{20} = \frac{u_2 g}{2k_2}$$

$$[k_{II}] = \begin{bmatrix} k_2 \Delta L_{20} 2L_1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$V_s = m_1 g h_{s1} + m_2 g h_{s2}$$

$$\begin{cases} h_{s1} = -L_1 \cos \vartheta_1 \\ h_{s2} = -2L_1 \cos \vartheta_1 + L_2 \sin \vartheta_2 \end{cases}$$

$$\frac{\partial^2 h_{s1}}{\partial \vartheta_1^2} = L_1 \cos \vartheta_1$$

$$\frac{\partial h_{s2}}{\partial \vartheta_1} = 2L_1 \sin \vartheta_1 ; \quad \frac{\partial^2 h_{s2}}{\partial \vartheta_1^2} = 2L_1 \cos \vartheta_1$$

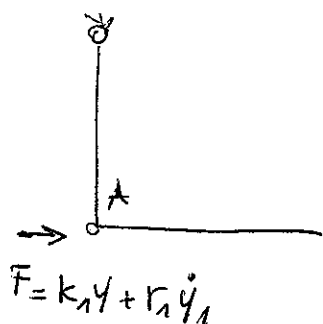
$$\frac{\partial h_{s2}}{\partial \vartheta_2} = L_2 \cos \vartheta_2 ; \quad \frac{\partial^2 h_{s2}}{\partial \vartheta_2^2} = -L_2 \sin \vartheta_2$$

$$\rightarrow [k_s] = \begin{bmatrix} m_1 g L_1 + 2m_2 g L_1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{aligned} \delta L &= \bar{F}_0 \cos \alpha \delta x_{s2} + \bar{F}_0 \sin \alpha \delta y_{s2} \\ &= \bar{F}_0 \cos \alpha 2L_1 \delta \vartheta_1 + \bar{F}_0 \sin \alpha L_2 \delta \vartheta_2 \end{aligned}$$

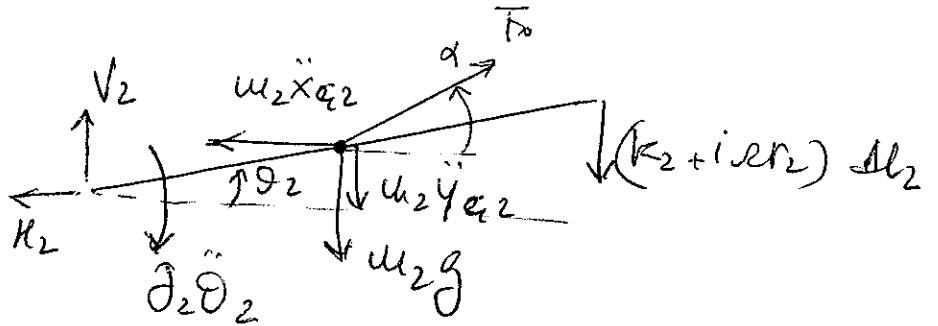
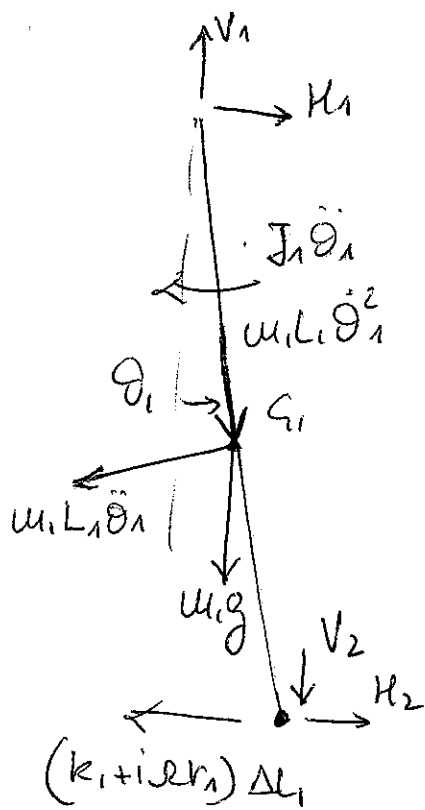
$$\rightarrow \underline{Q} = \begin{Bmatrix} 2L_1 \cos \alpha \\ L_2 \sin \alpha \end{Bmatrix} \bar{F}_0 e^{i\omega t}$$

Spostamento impresso:



$$\delta L = F \delta x_A = (k_1 + i\omega r_1) y_0 e^{i\omega t} 2L_1 \delta \vartheta_1$$

$$Q = \begin{Bmatrix} 2L_1 (k_1 + i\omega r_1) \\ 0 \end{Bmatrix} y_0 e^{i\omega t}$$



$$\sum F_v^{OA} = 0 : V_1 - m_1 g - \cancel{m_1 L_1 \dot{\theta}_1^2 \cos \theta_1} - m_1 L_1 \ddot{\theta}_1 \sin \theta_1 - V_2 = 0$$

$$\sum F_v^{AB} = 0 : V_2 - m_2 g - m_2 \ddot{y}_{s2} + \bar{T}_0 \sin \alpha - (k_2 + i r_2) \Delta l_2 = 0$$

$$V_2 = m_2 g + m_2 \ddot{y}_{s2} - \bar{T}_0 \sin \alpha + (k_2 + i r_2) \Delta l_2$$

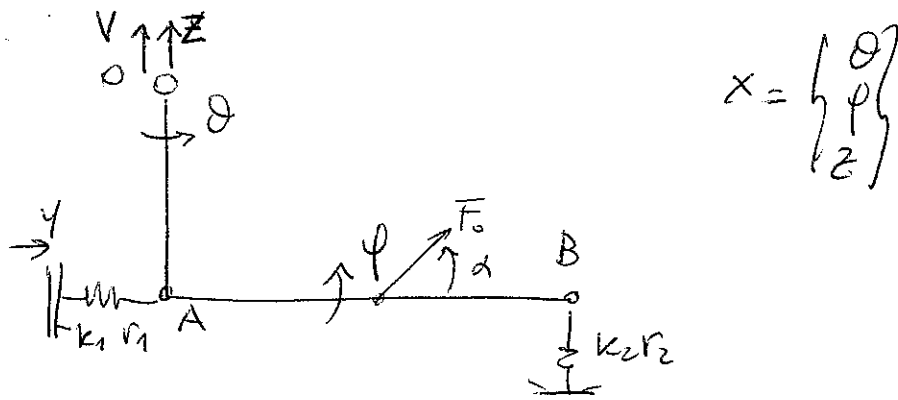
$$V_1 = m_1 g + \cancel{m_1 L_1 \dot{\theta}_1 \sin \theta_1} + m_2 g + m_2 \ddot{y}_{s2} + \bar{T}_0 \sin \alpha + (k_2 + i r_2) \Delta l_2$$

$$V_1 = m_1 g + m_2 g + m_2 \ddot{y}_{s2} + \bar{T}_0 \sin \alpha + (k_2 + i r_2) \Delta l_2$$

$$y_{s2} = L_2 \theta_2 \rightarrow \ddot{y}_{s2} = -\omega^2 L_2 \theta_2$$

$$\Delta l_2 = 2 L_2 \theta_2$$

$$V_{ADW} = m_2 \ddot{y}_{s2} - \bar{T}_0 \sin \alpha + (k_2 + i r_2) \Delta l_2$$



$$\theta = \phi = 0 \quad z \neq 0:$$

$$x_{s1} = 0$$

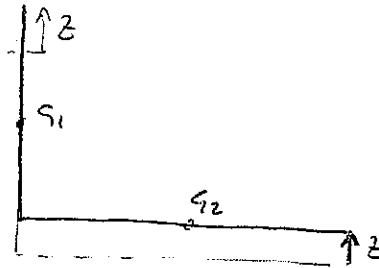
$$y_{s1} = z$$

$$x_{s2} = 0$$

$$y_{s2} = z$$

$$\Delta L_1 = 0$$

$$\Delta L_2 = z$$



$$u_F = \text{diag}(u_1, u_1, J_1, u_2, u_2, J_2)$$

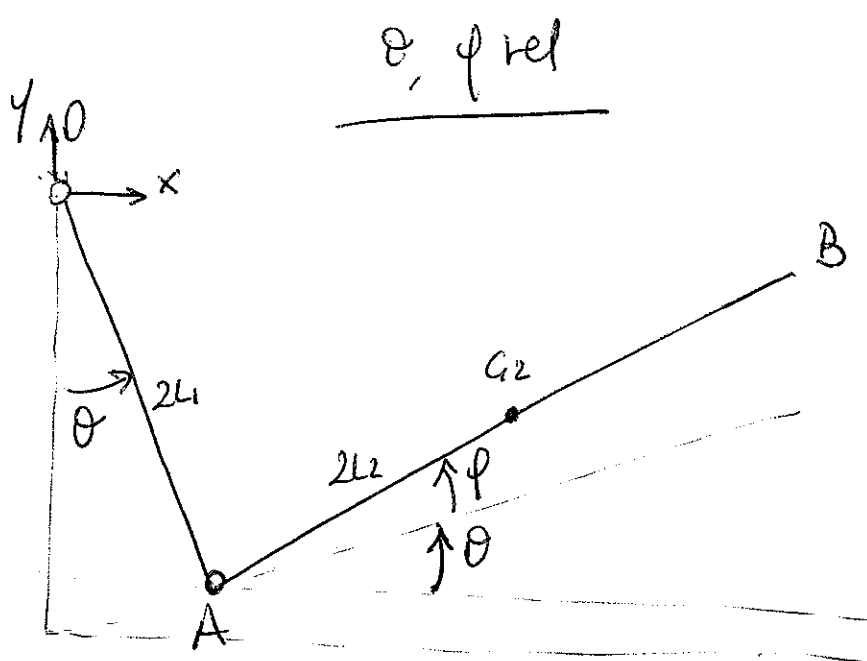
$$\Delta u = \begin{bmatrix} L_1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 2L_1 & 0 & 0 \\ 0 & L_2 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\Delta u = \begin{bmatrix} 2L_1 & 0 & 0 \\ 0 & 2L_2 & 1 \end{bmatrix}$$

$$\delta L = F_0 \cos \alpha \, 2L_1 \delta \theta + F_0 \sin \alpha \, L_2 \delta \phi + F_0 \sin \alpha \, \delta z + V \delta z$$

$$Q_L = \begin{Bmatrix} F_0 \cos \alpha \, 2L_1 \\ F_0 \sin \alpha \, L_2 \end{Bmatrix}$$

$$Q_V = \{ F_0 \sin \alpha + V \}$$



$$\begin{cases} x_{G2} = 2L_1 \sin \theta + L_2 \cos(\theta + \varphi) \\ y_{G2} = -2L_1 \cos \theta + L_2 \sin(\theta + \varphi) \end{cases} \rightarrow \begin{cases} x_{G2} \approx 2L_1 \theta + L_2 \\ y_{G2} \approx -2L_1 + L_2(\theta + \varphi) \end{cases}$$

$$\Delta L_2 = 2L_1 - 2L_1 \cos \theta + 2L_2 \sin(\theta + \varphi) \rightarrow \Delta L_2 \approx 2L_2(\theta + \varphi)$$

$$\frac{\partial \Delta L_2}{\partial \theta} = 2L_1 \sin \theta + 2L_2 \cos(\theta + \varphi)$$

$$\frac{\partial^2 \Delta L_2}{\partial \theta^2} = 2L_1 \cos \theta - 2L_2 \sin(\theta + \varphi) \rightarrow \left. \frac{\partial^2 \Delta L_2}{\partial \theta^2} \right|_0 = 2L_1$$

~~Matrix~~ $\Delta u = \begin{bmatrix} L_1 & 0 \\ 1 & 0 \\ 2L_1 & 0 \\ L_2 & L_2 \\ 1 & 1 \end{bmatrix} \quad [\Delta u] = \begin{bmatrix} 2L_1 & 0 \\ 2L_2 & 2L_2 \end{bmatrix}$

$$[k_q] = \begin{bmatrix} m_1 g L_1 + 2m_2 g L_1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$[k_{\bar{u}}] = \begin{bmatrix} 2k_2 L_1 \Delta L_2 & 0 \\ 0 & 0 \end{bmatrix}$$

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